Which Supports Are Settled

A machine-checked map for Erdős Problem #257, and a parity obstruction that closes the squarefree support to one class of argument

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Abstract

Erdős #257 asks whether $X_A = \sum_{a \in A}(2^a - 1)^{-1}$ is irrational for every infinite $A \subseteq \mathbb{N}_{>0}$. It is open. In December 2025 the question changed character: Tao and Teräväinen proved the case $A = \{\text{primes}\}$ unconditionally, remarked that the same method gives the prime powers, and wrote that it seems likely to reach other sets "sufficiently similar to the primes", leaving open which. The useful question is therefore now *which supports the available methods reach*.

This note answers that for the methods in this development. Section 2 gives a status map: seven infinite-support families are irrational in *every* integer base $b \geq 2$, each with a machine-checked proof, and none of the seven is reached by the analytic method, nor is the prime support reached by ours. Section 4 records the achievement-set geometry, and Section 5 the exact half-value dichotomy.

Section 3 contributes a limitation rather than a family. For the squarefree support the divisor incidence is exactly $2^{\omega(n)} - 1$, hence odd at every $n \geq 2$. Every certificate argument in this development opens by selecting a block with even incidence, so all of them fail on that one support — uniformly in the block, not for want of searching. This says nothing about whether the squarefree value is irrational, which is open. It says that one natural dense support sits outside both currently available methods, and names why.

Status. The problem treated here is open, and this note does not close it. Every statement below marked as checked is a proposition that the pinned Lean kernel accepts from the sources this note links to, with no `sorry`, no added axiom, and no unchecked evaluation. That is a claim about the formal statement, not about its mathematical interest, its novelty, or the original problem. The unresolved obligations are named exactly, in their own section, and none of the finite computations, reductions, or no-go results here removes one of them.

1 The problem

Problem 1.1 (Erdős #257)**.** For every infinite $A \subseteq \mathbb{N}_{>0}$, is $X_A = \sum_{a \in A} \dfrac{1}{2^a - 1}$ irrational?

Authorship and AI use. The prose in this version was generated by large-language-model agents under Will Cook's direction. Cook set the objectives and acceptance criteria, reviewed and authorised the manuscript, and is responsible for its contents. The agents are production tools, not authors. See *Statements and declarations*.

Numbering and status follow Bloom's catalogue. Write $w_n = (2^n - 1)^{-1}$. Interchanging sums gives the coordinate this note works in: with the *divisor incidence* $\mathrm{sc}_A(n) = \#\{d \mid n : d \in A\}$,

$$X_A = \sum_{a \in A} \frac{1}{2^a - 1} = \sum_{n \geq 1} \frac{\mathrm{sc}_A(n)}{2^n}. \tag{1}$$

So #257 is the question of which 0/1 weights make a Lambert series irrational, and every argument below reads the coefficient sequence sc_A rather than A itself.

2 The map of settled supports

Support A	Bases	Authority
all of $\mathbb{N}_{>0}$	every $b \geq 2$	Erdős 1948; checked here
primes	$b = 2$ (and $b \geq 2$)	Tao–Teräväinen [1]; not formalised here
prime powers	$b = 2$	*ibid.*, stated as a remark; not formalised here
factorials $\{n!\}$	every $b \geq 2$	checked here
powers of two $\{2^k\}$	every $b \geq 2$	checked here
multiples of a fixed d	every $b \geq 2$	checked here
a residue class	every $b \geq 2$	checked here
the odd numbers	every $b \geq 2$	checked here
eventually periodic	every $b \geq 2$	checked here
pairwise coprime, $\sum_{a \in A} a^{-1} < \infty$	every $b \geq 2$	checked here
squarefree numbers	—	**Open**; outside the engines here (Section 3)
arbitrary infinite A	—	**Open** (Problem 1.1)

Three things about this table are worth stating plainly.

First, the families checked here are proved for *every* integer base $b \geq 2$, not only base 2; the general statement is the base-b full-support theorem and the family results specialise it. Prior art is acknowledged rather than absorbed: for eventually periodic *rational* coefficient sequences Luca and Tachiya prove a broader theorem by different methods [3], and no priority is claimed for the formalised versions.

Second, the two method families are *complementary, not nested*. The analytic method of [1] reaches the primes and prime powers, which are neither eventually periodic nor pairwise coprime with summable reciprocals, so they are outside every row proved here. The rows proved here — factorials, powers of two, residue classes — are not "sufficiently similar to the primes" in the sense that method needs. Neither list contains the other, and their union does not exhaust the infinite supports.

Third, none of this approaches the universal statement. A list of settled families, however long, is a list; Problem 1.1 quantifies over all infinite A, and the gap between the two is not narrowed by adding rows.

3 A support outside both methods

Let $A_{\mathrm{sf}} = \{d \geq 2 : d \text{ squarefree}\}$. This is a natural support of density $6/\pi^2$ — far denser than the primes, and not periodic.

Theorem 3.1 (squarefree divisor incidence)**.** *For every $n \geq 1$ the number of squarefree divisors of n is $2^{\omega(n)}$, where $\omega(n)$ is the number of distinct prime factors, and hence*

$$\mathrm{sc}_{A_{\mathrm{sf}}}(n) = 2^{\omega(n)} - 1.$$

In particular $\mathrm{sc}_{A_{\mathrm{sf}}}(n)$ *is* odd *for every $n \geq 2$.*

The count is checked, the incidence formula is checked, and the parity conclusion is checked. The proof is the bijection between squarefree divisors of n and subsets of its prime factors (Lean), so the count is $2^{\omega(n)}$; removing $d = 1$ leaves an odd number whenever $\omega(n) \geq 1$.

Corollary 3.2 (the engines cannot see this support)**.** *Every certificate construction in this development opens by selecting a block position N with $2 \mid \mathrm{sc}_A(N+1)$. By Theorem 3.1 no such N exists for A_{sf}. The obstruction is uniform in N: it is not that a search has not gone far enough, but that the required residue never occurs.*

We are deliberate about what this does and does not say.

It does not say that $X_{A_{\mathrm{sf}}} = \sum_{d \text{ squarefree}, d\geq 2}(2^d - 1)^{-1}$ is rational, or that it is hard, or that it is irrational. That value's status is open and this note does not move it.

It does say that a specific, fully specified proof method provably cannot certify one natural support, for a reason internal to the support rather than to the search. Note also that the obstruction is a parity accident rather than a scaling one: the shifted coefficient $2^{\omega(n)}$ still satisfies $2^{\omega(n)} \leq n$ (checked), so it remains inside the coefficient range those arguments require. Repairing the parity is therefore not obviously impossible — but a repaired first block does not discharge the remaining tail and head obligations, and we make no claim that it does.

4 Achievement-set geometry

The set of all values obtainable from the base-2 Mersenne weights is the achievement set $\mathcal{A} = \{\sum_{n\geq 1} \varepsilon_n w_n : \varepsilon_n \in \{0, 1\}\}$.

Theorem 4.1 (geometry of $\mathcal{A}$)**.** *$\mathcal{A}$ is compact, perfect, totally disconnected and nowhere dense, and has Lebesgue measure* 1*.*

Checked as perfect, totally disconnected, nowhere dense, and of measure one. The strict-tail Cantor geometry is recorded by Kovač and Tao; no novelty is claimed for the formalised refinements. Membership is equivalent to survival of the greedy expansion at every level (Lean), and exact rational death certificates prove non-membership: $3/4 \notin \mathcal{A}$ (Lean).

5 The half-value dichotomy

A rational point of $\mathcal{A}$ attained by an infinite support would refute Problem 1.1. The distinguished candidate is $1/2$.

Theorem 5.1 (exact dichotomy at $1/2$)**.** *$1/2 \in \mathcal{A}$ if and only if the canonical greedy expansion of $1/2$ omits infinitely many exponents. Moreover no finite support has value $1/2$, so membership would automatically produce an* infinite *support of rational value, and hence a counterexample to universal #257.*

The equivalence is checked, and the finite-support exclusion is checked.

This is an equivalence, and we mark it as such rather than as progress: both sides are open, and proving them equivalent does not decide either. It is included because it fixes what a counterexample would have to look like, and because the negative branch is genuinely one-sided — non-membership is witnessed by a finite fatal gap and is therefore semi-decidable, whereas membership is an infinite conjunction that no finite computation certifies. Computing the greedy orbit further can only ever raise a lower bound on where death could occur; it cannot establish survival.

6 What remains open

1. Problem 1.1 itself, for arbitrary infinite A. The map in Section 2 is a list of families and does not approach a universal statement.
2. Irrationality of $X_{A_{\mathrm{sf}}}$ for the squarefree support. Section 3 closes one method against it and supplies no replacement.
3. Membership of $1/2$ in $\mathcal{A}$. Theorem 5.1 makes it exactly equivalent to infinitely many greedy skips; neither side is proved.
4. Which supports the analytic method of [1] reaches. The authors name this themselves and do not pursue it; it is, in our view, the most valuable open question in the immediate neighbourhood of #257, and it is not one this development is currently equipped to answer.

Statements and declarations

This manuscript is authored exposition, not proof authority. The linked Lean snapshot is authoritative only for its exact propositions; kernel checking establishes that a proposition was proved, not that it is interesting, novel, or sufficient. Results attributed to Tao and Teräväinen, to Luca and Tachiya, and to Kovač and Tao are cited from the literature and are not formalised here. Erdős Problem #257 remains open.

References

[1] T. Tao and J. Teräväinen, *Quantitative correlations and some problems on prime factors of consecutive integers*, arXiv:2512.01739 (submitted December 2025, revised April 2026). Proves $\sum_{n\geq 1} \omega(n)/2^n = \sum_p (2^p - 1)^{-1}$ irrational, settling the prime-support

case of #257; remarks that the method also gives the prime-power support and every integer base $b \geq 2$.

[2] K. Pratt, *The irrationality of a prime factor series under a prime tuples conjecture*, arXiv:2409.15185.

[3] F. Luca and Y. Tachiya, *Irrationality of Lambert series associated with a periodic sequence*.

[4] P. Erdős, *On arithmetical properties of Lambert series*, J. Indian Math. Soc. 12 (1948), 63–66.

[5] T. Bloom, *Erdős Problems*, https://www.erdosproblems.com/257.